

Exercise 2.7.12. Write $T^{(k)}$ for the k -fold Cartesian product $T \times \cdots \times T$. Prove⁽²⁹⁾ that $T^{(k)}$ is ergodic for all $k \geq 2$ if and only if $T^{(2)}$ is ergodic.

Exercise 2.7.13. Let T be an ergodic endomorphism of $\mathbb{T}^d$. The following exponential error rate for the mixing property⁽³⁰⁾,

$$\left| \langle f_1, U_T^n f_2 \rangle - \int f_1 \int f_2 \right| \leq S(f_1)S(f_2)\theta^n$$

for some $\theta < 1$ depending on T and for a pair of constants $S(f_1), S(f_2)$ depending on $f_1, f_2 \in C^\infty(\mathbb{T}^d)$, is known to hold.

- (a) Prove an exponential rate of mixing for the map $T_n : \mathbb{T} \rightarrow \mathbb{T}$ defined by $T_n(x) = nx \pmod{1}$.
- (b) Prove an exponential rate of mixing for the automorphism of $\mathbb{T}^2$ defined by $T : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} y \\ x+y \end{pmatrix}$.
- (c) Could an exponential rate of mixing hold for all continuous functions?

2.8 Proof of Weak-Mixing Equivalences

Some of the implications in Theorem 2.36 require the development of additional material; after developing it we will end this section with a proof of Theorem 2.36. The first lemma needed is a general one from analysis, due to Koopman and von Neumann [209].

Lemma 2.41. *Let (a_n) be a bounded sequence of non-negative real numbers. Then the following are equivalent:*

- (1) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} a_j = 0$;
- (2) *there is a set $J = J((a_n)) \subseteq \mathbb{N}$ with density zero for which $a_n \xrightarrow{n \notin J} 0$;*
- (3) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} a_j^2 = 0$.

PROOF. (1) $\implies$ (2): Let $J_k = \{j \in \mathbb{N} \mid a_j > \frac{1}{k}\}$, so that

$$J_1 \subseteq J_2 \subseteq J_3 \subseteq \cdots \quad (2.32)$$

For each $k \geq 1$,

$$\frac{1}{k} |J_k \cap [0, n)| < \sum_{\substack{i=0, \dots, n-1, \\ a_i > 1/k}} a_i \leq \sum_{i=0}^{n-1} a_i.$$

It follows that

$$\frac{1}{n} |J_k \cap [0, n)| \leq k \frac{1}{n} \sum_{i=0}^{n-1} a_i \longrightarrow 0$$

as $n \rightarrow \infty$ for each $k \geq 1$, so each J_k has zero density. We will construct the set J by taking a union of segments of each set J_k . Since each of the sets J_k has zero density, we may inductively choose numbers $0 < \ell_1 < \ell_2 < \dots$ with the property that

$$\frac{1}{n} |J_k \cap [0, n)| \leq \frac{1}{k} \quad (2.33)$$

for $n \geq \ell_k$ and any $k \geq 1$. Define the set J by

$$J = \bigcup_{k=0}^{\infty} (J_k \cap [\ell_k, \ell_{k+1})).$$

We claim two properties for the set J , namely

- $a_n \xrightarrow[n \notin J]{} 0$ as $n \rightarrow \infty$;
- J has density zero.

For the first claim, note that $J_k \cap [\ell_k, \infty) \subseteq J$ by (2.32), so if $J \not\ni n \geq \ell_k$ then $n \notin J_k$, and so $a_n \leq \frac{1}{k}$. This shows that $a_n \xrightarrow[n \notin J]{} 0$ as claimed.

For the second claim, notice that if $n \in [\ell_k, \ell_{k+1})$ then again by (2.32) $J \cap [0, n) \subseteq J_k \cap [0, n)$ and so

$$\frac{1}{n} |J \cap [0, n)| \leq \frac{1}{k}$$

by (2.33), showing that J has density zero.

(2) $\implies$ (1): The sequence (a_n) is bounded, so there is some $R > 0$ with $a_n \leq R$ for all $n \geq 1$. For each $k \geq 1$ choose N_k so that

$$J \not\ni n \geq N_k \implies a_n < \frac{1}{k}$$

and so that

$$n \geq N_k \implies \frac{1}{n} |J \cap [0, n)| \leq \frac{1}{k}.$$

Then for $n \geq kN_k$,

$$\begin{aligned} \frac{1}{n} \sum_{i=0}^{n-1} a_i &= \frac{1}{n} \left(\sum_{i=0}^{N_k-1} a_i + \sum_{\substack{i \in J, \\ N_k \leq i < n}} a_i + \sum_{\substack{i \notin J, \\ N_k \leq i < n}} a_i \right) \\ &< \frac{1}{n} \left(RN_k + R|J \cap [0, n)| + n \frac{1}{k} \right) \\ &\leq \frac{2R+1}{k}, \end{aligned}$$

showing (1).

(3) $\iff$ (1): This is clear from the characterization (2) of property (1). $\square$

PROOF OF THEOREM 2.36. Properties (1), (6) and (7) are equivalent by Lemma 2.41 applied with $a_n = |\mu(A \cap T^{-n}B) - \mu(A)\mu(B)|$.

(6) $\implies$ (3): Given sets $A_1, B_1, A_2, B_2 \in \mathcal{B}$, property (6) gives sets J_1 and J_2 of density zero with

$$\mu(A_1 \cap T^{-n}B_1) \xrightarrow{n \notin J_1} \mu(A_1)\mu(B_1)$$

and

$$\mu(A_2 \cap T^{-n}B_2) \xrightarrow{n \notin J_2} \mu(A_2)\mu(B_2).$$

Let $J = J_1 \cup J_2$; this still has density zero and

$$\begin{aligned} \lim_{J \not\ni n \rightarrow \infty} & \left| (\mu \times \mu) \left((A_1 \times A_2) \cap (T \times T)^{-n} (B_1 \times B_2) \right) \right. \\ & \quad \left. - (\mu \times \mu)(A_1 \times A_2) \cdot (\mu \times \mu)(B_1 \times B_2) \right| \\ &= \lim_{J \not\ni n \rightarrow \infty} \left| \mu(A_1 \cap T^{-n}B_1) \cdot \mu(A_2 \cap T^{-n}B_2) \right. \\ & \quad \left. - \mu(A_1)\mu(A_2)\mu(B_1)\mu(B_2) \right| \\ &= 0, \end{aligned}$$

so $T \times T$ is weak-mixing since the measurable rectangles generate $\mathcal{B} \times \mathcal{B}$.

(3) $\implies$ (1): If $T \times T$ is weak-mixing, then property (1) holds in particular for subsets of $X \times X$ of the form $A \times X$ and $B \times X$, which shows that (1) holds for T , so T is weak-mixing.

(1) $\implies$ (4): Let $(Y, \mathcal{B}_Y, \nu, S)$ be an ergodic system and assume that T is weak-mixing. For measurable sets $A_1, B_1 \in \mathcal{B}$ and $A_2, B_2 \in \mathcal{B}_Y$,

$$\begin{aligned} & \frac{1}{N} \sum_{n=0}^{N-1} (\mu \times \nu) \left((A_1 \times A_2) \cap (T \times S)^{-n} (B_1 \times B_2) \right) \\ &= \frac{1}{N} \sum_{n=0}^{N-1} \mu(A_1 \cap T^{-n}B_1) \nu(A_2 \cap S^{-n}B_2) \\ &= \frac{1}{N} \sum_{n=0}^{N-1} \mu(A_1) \mu(B_1) \nu(A_2 \cap S^{-n}B_2) \\ & \quad + \frac{1}{N} \sum_{n=0}^{N-1} [\mu(A_1 \cap T^{-n}B_1) - \mu(A_1)\mu(B_1)] \nu(A_2 \cap S^{-n}B_2). \quad (2.34) \end{aligned}$$

By the characterization in (2.31) and ergodicity of S , the expression on the right in (2.34) converges to

$$\mu(A_1)\mu(B_1)\nu(A_2)\nu(B_2).$$

The second term in (2.34) is dominated by

$$\frac{1}{N} \sum_{n=0}^{N-1} |\mu(A_1 \cap T^{-n}B_1) - \mu(A_1)\mu(B_1)|$$

which converges to 0 since T is weak-mixing. It follows that

$$\frac{1}{N} \sum_{n=0}^{N-1} (\mu \times \nu)(A_1 \times A_2 \cap (T \times S)^{-n}(B_1 \times B_2)) \longrightarrow \mu(A_1)\mu(B_1)\nu(A_2)\nu(B_2)$$

so $T \times S$ is ergodic by the characterization in (2.31).

(4) $\implies$ (2): Let $(Y, \mathcal{B}_Y, \nu, S)$ be the ergodic system defined by the identity map on the singleton $Y = \{y\}$. Then $T \times S$ is isomorphic to T , so (4) shows that T is ergodic. Invoking (4) again now shows that $T \times T$ is ergodic, proving (2).

(2) $\implies$ (7): We must show that

$$\frac{1}{N} \sum_{n=0}^{N-1} |\mu(A \cap T^{-n}B) - \mu(A)\mu(B)|^2 \longrightarrow 0$$

as $N \rightarrow \infty$, for every $A, B \in \mathcal{B}$. Let μ^2 denote the product measure $\mu \times \mu$ on $(X \times X, \mathcal{B} \otimes \mathcal{B})$. By the ergodicity of $T \times T$,

$$\begin{aligned} \frac{1}{N} \sum_{n=0}^{N-1} \mu(A \cap T^{-n}B) &= \frac{1}{N} \sum_{n=0}^{N-1} \mu^2((A \times X) \cap (T \times T)^{-n}(B \times X)) \\ &\longrightarrow \mu^2(A \times X) \cdot \mu^2(B \times X) = \mu(A)\mu(B) \end{aligned}$$

and

$$\begin{aligned} \frac{1}{N} \sum_{n=0}^{N-1} (\mu(A \cap T^{-n}B))^2 &= \frac{1}{N} \sum_{n=0}^{N-1} \mu^2((A \times A) \cap (T \times T)^{-n}(B \times B)) \\ &\longrightarrow \mu^2(A \times A) \cdot \mu^2(B \times B) = \mu(A)^2 \mu(B)^2. \end{aligned}$$

It follows that

$$\begin{aligned}
\frac{1}{N} \sum_{n=0}^{N-1} [\mu(A \cap T^{-n}B) - \mu(A)\mu(B)]^2 &= \frac{1}{N} \sum_{n=0}^{N-1} \mu(A \cap T^{-n}B)^2 \\
&\quad + \mu(A)^2 \mu(B)^2 \\
&\quad - 2\mu(A)\mu(B) \frac{1}{N} \sum_{n=0}^{N-1} \mu(A \cap T^{-n}B) \\
&\rightarrow 2\mu(A)^2 \mu(B)^2 - 2\mu(A)^2 \mu(B)^2 = 0,
\end{aligned}$$

so (7) holds.

(2) $\implies$ (5): Suppose that f is a measurable eigenfunction for T , so

$$U_T f = \lambda f$$

for some $\lambda \in \mathbb{S}^1$. Define a measurable function on $X \times X$ by

$$g(x_1, x_2) = f(x_1) \overline{f(x_2)};$$

then

$$U_{T \times T} g(x, y) = g(Tx, Ty) = \lambda \bar{\lambda} g(x, y) = g(x, y)$$

so by ergodicity of $T \times T$, g (and hence f) must be constant almost everywhere.

All that remains is to prove that (5) $\implies$ (2), and this is considerably more difficult. There are several different proofs, each of which uses a non-trivial result from functional analysis⁽³¹⁾. Assume that $T \times T$ is not ergodic, so there is a non-constant function $f \in L^2_{\mu^2}(X \times X)$ that is almost everywhere invariant under $T \times T$. We would like to have the additional symmetry property $f(x, y) = \overline{f(y, x)}$ for all $(x, y) \in X \times X$. To obtain this additional property, consider the functions

$$(x, y) \mapsto f(x, y) + \overline{f(y, x)}$$

and

$$(x, y) \mapsto i(f(x, y) - \overline{f(y, x)}).$$

Notice that if both of these functions are constant, then f must be constant. It follows that one of them must be non-constant. So without loss of generality we may assume that f satisfies $f(x, y) = \overline{f(y, x)}$. We may further suppose (by subtracting $\int f d\mu^2$) that $\int f d\mu^2 = 0$. It follows that the operator F on L^2_{μ} defined by

$$(F(g))(x) = \int_X f(x, y) g(y) d\mu(y)$$

is a non-trivial self-adjoint compact⁽³²⁾ operator, and so by Theorem B.3 has at least one non-zero eigenvalue λ whose corresponding eigenspace V_λ is finite-dimensional. We claim that the finite-dimensional space $V_\lambda \subseteq L_\mu^2$ is invariant under T . To see this, assume that $F(g) = \lambda g$. Then

$$\begin{aligned} \lambda g(Tx) &= \int_X f(Tx, y)g(y) \, d\mu(y) \\ &= \int_X f(Tx, Ty)g(Ty) \, d\mu(y) \quad (\text{since } \mu \text{ is } T\text{-invariant}) \\ &= \int_X f(x, y)g(Ty) \, d\mu(y), \end{aligned}$$

since f is $T \times T$ -invariant, so $F(g \circ T) = \lambda(g \circ T)$ and thus $g \circ T \in V_\lambda$. It follows that U_T restricted to V_λ is a non-trivial linear map of a finite-dimensional linear space, and therefore has a non-trivial eigenvector. Since $\int f \, d\mu^2 = 0$, any such eigenvector is non-constant. $\square$

2.8.1 Continuous Spectrum and Weak-Mixing

A more conventional proof of the difficult step in Theorem 2.36, which may be taken to be (5) $\implies$ (1), proceeds via the Spectral theorem (Theorem B.4) in the following form.

ALTERNATIVE PROOF OF (5) $\implies$ (1) IN THEOREM 2.36. Definition 2.35 is clearly equivalent to the property that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} |\langle U_T^n f, g \rangle - \langle f, 1 \rangle \cdot \langle 1, g \rangle| = 0$$

for any $f, g \in L_\mu^2$, and by polarization this is in turn equivalent to

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} |\langle U_T^n f, f \rangle - \langle f, 1 \rangle \cdot \langle 1, f \rangle| = 0$$

for any $f \in L_\mu^2$ (see Exercise 2.7.8 and page 441). By subtracting $\int_X f \, d\mu$ from f , it is therefore enough to show that if $f \in L_\mu^2$ has $\int_X f \, d\mu = 0$, then

$$\frac{1}{N} \sum_{n=0}^{N-1} |\langle U_T^n f, f \rangle|^2 \longrightarrow 0$$

as $N \rightarrow \infty$. By (B.1), it is enough to show that for the non-atomic measure μ_f on $\mathbb{S}^1$,

$$\frac{1}{N} \sum_{n=0}^{N-1} \left| \int_{\mathbb{S}^1} z^n d\mu_f(z) \right|^2 \longrightarrow 0 \quad (2.35)$$

as $N \rightarrow \infty$. Since $\overline{z^n} = z^{-n}$ for $z \in \mathbb{S}^1$ the product in (2.35) may be expanded to give

$$\begin{aligned} \frac{1}{N} \sum_{n=0}^{N-1} \left| \int_{\mathbb{S}^1} z^n d\mu_f(z) \right|^2 &= \frac{1}{N} \sum_{n=0}^{N-1} \left(\int_{\mathbb{S}^1} z^n d\mu_f(z) \cdot \int_{\mathbb{S}^1} w^{-n} d\mu_f(w) \right) \\ &= \frac{1}{N} \sum_{n=0}^{N-1} \int_{\mathbb{S}^1 \times \mathbb{S}^1} (z/w)^n d\mu_f^2(z, w) \quad (\text{by Fubini}) \\ &= \int_{\mathbb{S}^1 \times \mathbb{S}^1} \left(\frac{1}{N} \sum_{n=0}^{N-1} (z/w)^n \right) d\mu_f^2(z, w). \end{aligned}$$

The measure μ_f is non-atomic so the diagonal set $\{(z, z) \mid z \in \mathbb{S}^1\} \subseteq \mathbb{S}^1 \times \mathbb{S}^1$ has zero μ_f^2 -measure. For $z \neq w$,

$$\frac{1}{N} \sum_{n=0}^{N-1} (z/w)^n = \frac{1}{N} \left(\frac{1 - (z/w)^N}{1 - (z/w)} \right) \longrightarrow 0$$

as $N \rightarrow \infty$, so the convergence (2.35) holds by the dominated convergence theorem (Theorem A.18). $\square$

Exercises for Sect. 2.8

Exercise 2.8.1. Is the hypothesis that the sequence (a_n) be bounded necessary in Lemma 2.41?

Exercise 2.8.2. Give an alternative proof of (1) $\implies$ (5) in Theorem 2.36 by proving the following statements:

- (1) Any factor of a weak-mixing transformation is weak-mixing.
- (2) A complex-valued eigenfunction f of U_T has constant modulus.
- (3) If f is an eigenfunction of U_T , then $x \mapsto \arg(f(x)/|f(x)|)$ is a factor map from $(X, \mathcal{B}, \mu, T)$ to $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}}, R_{\alpha})$ for some α .

Exercise 2.8.3. Show the following converse to Exercise 2.5.6: if a measure-preserving system $(Y, \mathcal{B}_Y, \nu, S)$ is not totally ergodic then there exists a measure-preserving system $(X, \mathcal{B}, \mu, T)$ and a $K > 1$ with the property that $(Y, \mathcal{B}_Y, \nu, S)$ is measurably isomorphic to the system

$$(X^{(K)}, \mathcal{B}^{(K)}, \mu^{(K)}, T^{(K)})$$

constructed in Exercise 2.5.6.